

Making a Guinea Pig Monster in *The Bad Guys*

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Figure 1: Guinea Pig Final Shots

ABSTRACT

In the film *The Bad Guys*, the villain uses a mind control device to manipulate an army of guinea pigs which in themselves evolve through various stages of scale and coalescence, ultimately forming a large monster. In order to create congruity between these different scales of motion we created targeted solutions and worked cross departmentally in an effort to maintain artistic control throughout the process.

CCS CONCEPTS

• Computing methodologies → Computer graphics.

KEYWORDS

simulation, fluid, dynamics

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1 DESIGN/COLLABORATION

Creating a monster at a massive scale while retaining the cartoony and stylized nature of our film proved challenging and required

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multiple departments working together to process specific aspects of the design and motion of the creature. Animation drove overall performance and choreography of the monster via a "sock puppet" rig, which served as a target shape for simulations of the mass. Crowds provided small-scale details such as a leading front-edge and individual runners. During the simulation stage emphasis was placed on maintaining traditional animation and graphic design components such as silhouettes, snappy timings and level of detail shifts.

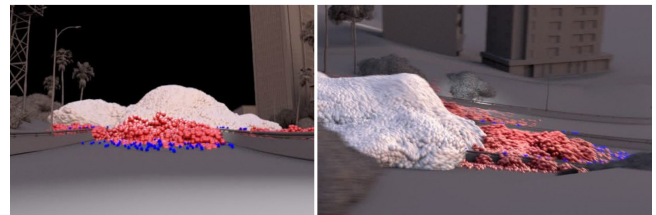


Figure 2: Rolling rig in red, large scale simulation in white, crowd simulation in blue

2 SMALL SCALE ROLLING RIG

This was achieved by creating a procedural rolling point cloud that would then be run through a position based dynamics solver which allowed us to set target positions creatively while still being able to solve for collisions. Transforming the rig would automatically rotate the mass much like a tire wheel and then artists could creatively lattice deform the shape into many stylized variations. We could resolve any post deform issues by resampling the data set to be more uniform in density and then added galloping by using a procedural

sine wave function. This was all done prior to any simulation step and could be turned around quickly to approximate the final result. A subsequent solve would then add more organic details and resolve any collisions. The target stiffness coefficient could be dialed per shot and used for adding more organic features, such as outer edge lag and clumping. This simulation step also grounded the shape since gravity would push any guinea pig to the lowest possible point in the mass and gave us the illusion of characters climbing on top of one another.

The rolling rig was used for small to medium scale details where we wanted to see individual guinea pigs crawling on top of one another and was often used as a bridge between the main mass and individual guinea pigs. It could be dressed and positioned separately which gave us the ability to collaborate more efficiently when designing these shots. When breaking off on its own, it took on its own unique animalistic galloping behaviour.



Figure 3: top left: animation, top right: bound points, bottom left: flow curves, bottom right: simulation

3 LARGE SCALE MONSTER SIMULATION

The main monster required an alternative solution due to its larger mass and faster scale. We used a FLIP based fluid solver with a quadratically interpolated target distance field as a suction force. A rest space bound point cloud would provide the velocity influence of the monster and the distance field would be used for solving the pressure. We could also dial the influence of the target based on the speed of the monster. For this effect we used a fairly low voxel grid scale which gave a less liquidy result but also gave the look of secondary animation or jiggle to the monster at the desired frequency level.

To drive the currents of flow in the monster we created two sets of curves that were procedurally generated from the base geometry. The first set of curves would drive motion on the body of the monster while the second set would create a set of legs that help ground the creature to the environment. These flow curves were created by finding high density point regions in the base animation geometry, then segmenting these points into start/end groups and using a shortest path method for tracing the curves along the geometry. The legs were achieved by creating points at the intersection of the monster and environment and then projecting these points out to the environment, tracing them to create lines and then procedurally controlling the line growth to reach out, attach and then suck back into the mass. To break up excess uniformity we also added individual wandering to each guinea pig in a pre-advection step.

A common issue was volume loss due to collisions with the environment and the constantly changing shape and volume of the underlying geometry. To solve this problem we created two volume loss compensation solutions. The first was to cut portions of the collision environment by intersecting it with an eroded volume of the underlying monster base geometry, and the other was to create a skeleton of points that roughly lived at the medial axis of the base geometry and could source new guinea pigs when overall volume started to decrease. Events and specific collision behaviours were handled separately for greater control.

4 POST PROCESSING/STYLIZATION

While solver flow overshoot provided aesthetically-pleasing secondary motion, there were situations where we wanted the resulting guinea pig mass to exactly match the sock puppet animation. Specifically any time the monster was swiping and near missing the characters. In addition, we also wanted to occasionally maintain the silhouettes established in the base sock puppet phase. For this we did a post simulation retargeting by calculating the distance to the original sock puppet bound point positions and using blend attributes to move the guinea pig mass back to its original state without impacting the flow behaviour of the simulation. Stylized poses and silhouettes were also hit by post-sim particle painting using our in house tools. Artists could pick a frame, paint the desired shape in camera and then control blending in and out of those poses without altering the nature of the simulation.

5 INSTALLATION AND RENDERING

For rendering our processed simulation we instanced various guinea pig animation cycles. Animators created a multitude of cycles to use, from standard running and walking, to some falling and more comical cycles. We based the cycle assignment for the monster on the proximity to any hard surface. Guinea pigs closer to the ground would typically be running, while guinea pigs within the monster would transition from walk to run based on internal velocity (global velocity of the monster minus local velocity of the simulation). Event based guinea pigs would transition from run to falling/spinning based on velocity as well.

Prior to beginning shot production, we created a procedural template of the complete workflow that would automatically run all the shots and would create a viewable sequence cut. This gave us the ability to quickly identify areas of concern and gave the entire production early access to simulated results which in turn helped produce a more cohesive and artistically defined final result.

Specific AOV's were generated for final lighting integration including per instance ID to randomize shading, level set depth to fade detail into the larger mass and reference position to define spatially coherent noise. AOV's were also used in conjunction with a Kuwahara anisotropic filter to reduce high frequency details into painterly strokes, properly seating our data within the overall style of the show.